

Report
Of
Legal/RTI Document Assistant using Corrective RAG & GraphRAG

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

Roll Number of Student: CS-23411301

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Batch: 2023-2027

CANDIDATE'S DECLARATION

I, NAND KISHOR KUMAR, do hereby declare that the internship report titled "Legal/RTI Document Assistant using Corrective RAG & GraphRAG" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: NAND KISHOR KUMAR

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program at Veda Tech. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

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INTERNSHIP COMPLETION CERTIFICATE

Vedas Tech

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INTERNSHIP COMPLETION CERTIFICATE

To whom it may concern

This is to certify that **Mr. Nand Kishor Kumar** has done his internship as an AI/ML Product Engineering Intern at **Vedas Tech** from **09/06/26- 10/08/26**.

He has worked on a live project and assisted the team in developing AI/ML-based solutions and tools. This project was aimed at enhancing automation and improving performance through data-driven models.

He has done extensive work on understanding the product, implementing machine learning models, and has learnt skills beyond his area of interest to achieve results for the company.

During the Internship he has shown exemplary hard work and has also been instrumental in getting the initial research and development work done and has completed all the tasks entrusted to him in time and in a good manner.

We wish him all the best for all his future goals and wishes.

For Vedas Tech

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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

This report presents an account of the internship undertaken at Veda Tech, where I served as an AI/ML Product Engineering Intern from 9th June 2026 to 10th August 2026 (2 months). During this internship, I developed an advanced Retrieval-Augmented Generation (RAG) system titled "Legal/RTI Document Assistant" — an AI-powered tool designed to provide citizens with accurate, citation-grounded answers to queries on RTI Acts, court judgments, and CIC decisions.

4.2 Organization Profile

Veda Tech is an AI and Language Services company operating since 2021, headquartered in Anand, Gujarat. The company focuses on AI-driven products and language-based solutions, and provides interns with hands-on, real-world product engineering experience.

4.3 Problem Statement

Citizens in India face significant difficulty in understanding the RTI Act, state-specific RTI rules, and related court judgments or CIC decisions due to the complexity of legal language and the time-consuming process of locating relevant precedents. There is a need for a system that can answer natural-language queries accurately, with proper citations, and without generating hallucinated (incorrect) information.

4.4 Project Objectives

- Index the RTI Act, State RTI Acts, court judgments, and CIC decisions in a structured manner
- Retrieve accurate answers with section/clause-level precision
- Generate hallucination-free, citation-grounded responses
- Link related legal precedents through graph-based relationships

4.5 Scope of the Project

The project currently covers the Central RTI Act 2005, the Delhi RTI Act 2001, Delhi High Court judgments, and CIC decisions. The system is designed to be extensible to additional state Acts and judgments in the future.

4.6 Technologies and Tools Used

Python, pdfplumber (PDF extraction), sentence-transformers (embeddings), rank-bm25 (keyword retrieval), Qdrant (vector database), Groq (LLM inference), python-dotenv, VS Code, and Git.

4.7 System Architecture

The system employs a parent-child chunking approach, where a parent chunk represents an entire Section and a child chunk represents an individual clause (e.g., Section 8(1)(j)). A hybrid retrieval mechanism (BM25 + vector search) surfaces relevant chunks, which then pass through a self-grading/re-retrieval loop. Following answer generation, a hallucination/faithfulness check is performed, and a citation graph layer links legal clauses to related case documents.

4.8 Methodology

1. Data collection: RTI Act 2005, Delhi RTI Act 2001, court judgments, and CIC decisions
2. Text extraction and cleaning (resolving OCR issues and font-mapping bugs)
3. Parent-child chunking with metadata (section, subsection, clause)
4. Indexing into the Qdrant vector database
5. Hybrid retrieval (BM25 + vector) and generation via Groq with citations
6. Addition of a hallucination/faithfulness verification layer with retry-on-fail logic
7. Development of a citation graph layer linking Act clauses to citing case documents
8. End-to-end testing (forward/reverse queries, no-graph fallback cases)

4.9 Expected Outcomes

A working Corrective RAG + GraphRAG pipeline capable of delivering accurate, citation-backed answers while automatically detecting and correcting hallucinated information. System accuracy was evaluated using RAGAS metrics (faithfulness, answer relevancy, context precision/recall).

4.10 Certificates of Completion and Other Communication Mail Proof

The offer/confirmation letter issued by Vedaas Tech is attached as proof of the internship. The Internship Completion Certificate will be attached upon successful completion of the internship period.

CHAPTER 5: BIBLIOGRAPHY / REFERENCES

- [1] Right to Information Act, 2005 (Government of India).
- [2] Delhi Right to Information Act, 2001 (Government of NCT of Delhi).
- [3] Rakesh Kumar Gupta v. Central Information Commission, Delhi High Court, 2021.
- [4] Prasanta Kumar Sahoo v. Ministry of Labour, Central Information Commission Decision, 2026.
- [5] Qdrant Documentation — Vector Database for AI Applications, <https://qdrant.tech/documentation/>
- [6] Sentence-Transformers Documentation, <https://www.sbert.net/>
- [7] Groq API Documentation, <https://groq.com/>
- [8] RAGAS: Evaluation Framework for Retrieval-Augmented Generation, <https://docs.ragas.io/>